HeatWise: Thermal-Aware Workload Placement for High-Density AI Data Centers Using a Reduced-Order Surrogate Model

Aryaman Pandey
Google DeepMind / HeatWise Research Team
Software Release Commit: 76fe215daee5462b55515045a2495c2725179fe6
Artifact Repository: https://github.com/Yurirain/Heatwise

Abstract—Modern artificial intelligence (AI) data centers face severe thermal challenges driven by extreme accelerator power densities and localized hot-aisle recirculation. Conventional cluster orchestrators—such as Round Robin and Least Loaded—prioritize compute resource availability while remaining thermally blind, frequently placing power-intensive workloads into localized thermal pockets and triggering severe hot-aisle recirculation. While full Computational Fluid Dynamics (CFD) simulations capture complex Navier-Stokes convective transport, their high computational cost (minutes to hours per state evaluation) prohibits their use in real-time cluster admission control.

We present HeatWise, a simulation-grounded, thermal-aware workload placement framework designed for high-density accelerator facilities. HeatWise combines an analytical lumped-parameter reduced-order thermal model with an empirical Random Forest surrogate model to predict post-placement rack equilibrium temperatures prior to physical dispatch. The surrogate operates on a strictly leakage-free 17-feature pre-placement schema and feeds a constrained multi-factor placement heuristic that optimizes thermal safety (35%), thermal headroom (25%), compute capacity fit (20%), local heat removal capacity (10%), and neighbor thermal stability (10%).

Across closed-loop dynamic trajectory evaluations against an analytical physical oracle over 250 sequential scheduling decisions, the production surrogate model achieves a dynamic Mean Absolute Error (MAE) of 1.867°C, a Top-1 placement agreement of 79.60%, a Top-3 overlap of 100.0%, a Spearman rank correlation (ρ) of 0.580, and a mean optimality gap of +0.473°C, with zero unsafe placements into candidate racks exceeding the 75.0°C critical threshold. In controlled multi-scheduler benchmark simulations under an invariant, decoupled workload stream, HeatWise reduces spatial cluster temperature variance to 8.9°C standard deviation (a 15.2% variance reduction relative to Round Robin at 10.5°C), while maintaining zero placement failures. Furthermore, serialized deployment (n_jobs=1) achieves a single-rack inference latency of 2.15 ms and cold artifact initialization of 23.64 ms, demonstrating practical viability for real-time cluster admission control.

Index Terms—Thermal-aware scheduling, AI accelerators, data center thermal management, machine learning surrogate models, reduced-order modeling, cluster resource management.

I. Introduction

The rapid scaling of deep neural network training and large language model (LLM) serving has driven data center power densities beyond 40–100 kW per rack [1]–[3]. Modern high-density compute nodes dissipate massive heat fluxes across compact form factors, creating steep localized thermal gradients and non-uniform hot-aisle recirculation patterns [4], [5].

Traditional cluster orchestrators (e.g., Kubernetes, SLURM, Google Borg [6]) schedule incoming compute tasks primarily based on static compute metrics: accelerator availability, memory headroom, and CPU utilization. This thermal blindness leads to severe operational inefficiencies:

1) Accelerated Hotspot Formation: Scheduling a burst of compute-heavy workloads onto a rack with degraded heat dissipation or elevated ambient recirculation triggers localized thermal hotspots exceeding critical thresholds ($T_{crit} = 75.0$°C).
2) Thermal SLA Violations: Operating temperatures climbing toward or beyond maximum safe limits ($T_{max_safe} = 80.0$°C) force accelerator thermal throttling, degrading application throughput and threatening hardware longevity [7].
3) Spatial Thermal Variance: Uneven heat distribution creates wide temperature differentials across the facility floor, forcing cooling infrastructure to over-provision airflow to satisfy the hottest node [8].

Full-physics Computational Fluid Dynamics (CFD) simulations accurately capture Navier-Stokes thermal transport but require minutes to hours per iteration [9], rendering them unusable for online cluster scheduling where admission decisions must resolve within milliseconds.

HeatWise addresses this fundamental gap through a surrogate modeling approach:

- We formulate an analytical lumped-parameter reduced-order thermal model capturing conductive, convective, and cross-rack recirculation physics across a 12-rack facility.
- We train an offline Random Forest Regressor surrogate [10] on 10,000 synthetic sequential transition vectors using an exact 17-feature leakage-free pre-placement schema.
- We develop a multi-factor placement engine that scores racks across thermal safety, headroom, compute fit, cooling capacity, and neighbor stability,

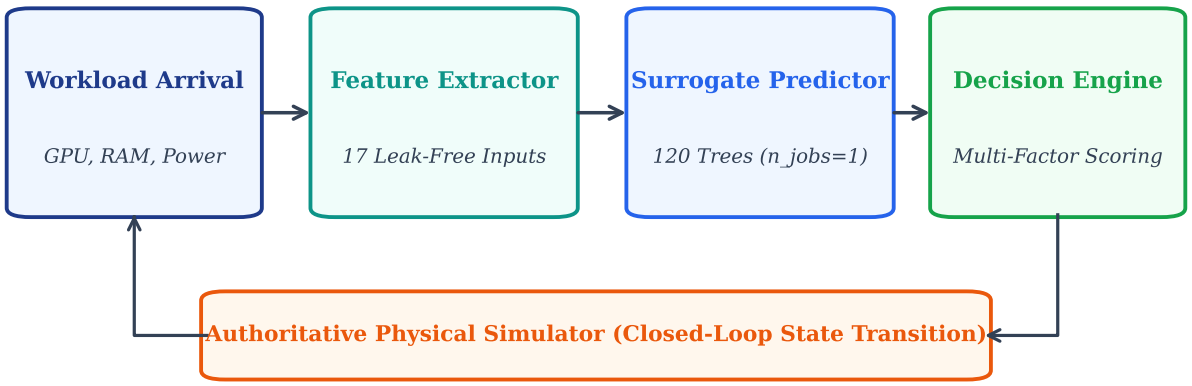


Fig. 1. End-to-end HeatWise system architecture. When an incoming workload arrives, the system extracts 17 pre-placement features, queries the Random Forest surrogate model to predict post-placement equilibrium temperatures across all 12 racks in 26.56 ms, applies multi-factor candidate filtering and scoring, dispatches the task, and feeds state updates into the physical environment.

enforcing strict safety thresholds ($T_{crit} = 75.0°C$, $T_{max_safe} = 80.0°C$).

The complete HeatWise pipeline is illustrated in Fig. 1. The software framework is available as an open-source artifact release [11]. The remainder of this paper is structured as follows. Section II reviews related work. Section III presents the simulated facility architecture. Section IV details the governing reduced-order thermal equations. Section V describes the dataset and feature engineering. Section VI presents the placement algorithm. Section VII outlines the experimental evaluation methodology. Section VIII and Section IX report static and dynamic ML validation results. Section X presents the multi-scheduler benchmark. Section XI details runtime performance and software verification. Section XII discusses limitations, and Section XIII concludes.

II. Related Work

Prior research on data center thermal management broadly partitions into three methodologies:

A. Reactive Thermal Throttling and DVFS

Hardware-level dynamic voltage and frequency scaling (DVFS) reduces clock speeds when internal silicon junctions reach critical temperatures. While reactive DVFS protects silicon from catastrophic thermal breakdown, it incurs unpredictable job latency spikes, straggler effects in distributed training, and fails to mitigate the hot-aisle recirculation patterns that cause the heating [5].

B. Computational Fluid Dynamics (CFD)

High-fidelity numerical fluid modeling solves the Navier-Stokes equations for air velocity, pressure, and temperature fields across data center geometries [9]. While CFD models provide millimeter-scale spatial fidelity and accurately reflect complex convective recirculation, each simulation step consumes hundreds of CPU-seconds to hours. This computational burden restricts CFD strictly to offline facility design, airflow containment planning, and post-hoc thermal auditing, making it inapplicable for real-time task admission.

C. Machine Learning Surrogates in Thermal Management

To bridge the latency gap between CFD and online schedulers, data-driven surrogate models have gained traction [4], [12]. Early efforts leveraged linear regression, multi-layer perceptrons, or support vector machines to estimate steady-state temperatures from CPU utilization. However, existing literature reveals two pervasive shortcomings:

1) Temporal Feature Leakage: Many published models include post-dispatch telemetry (e.g., measured post-placement outlet temperature or actual post-execution power draw) in their feature representations, yielding deceptively high offline R^2 scores that collapse when evaluated online.
2) Absence of Closed-Loop Dynamic Validation: Surrogates are predominantly validated on static, independently and identically distributed (i.i.d.) test splits. Under closed-loop execution, where model decisions directly alter future physical states, cumulative error drift can rapidly degrade placement quality.

HeatWise addresses both challenges by enforcing a strict pre-placement feature schema and validating performance across multi-step closed-loop trajectories against an analytical physical oracle.

III. Facility Topology and System Architecture

The simulated data center represents an enterprise AI accelerator facility organized as a 3×4 physical grid of 12 high-density racks, illustrated in Fig. 2.

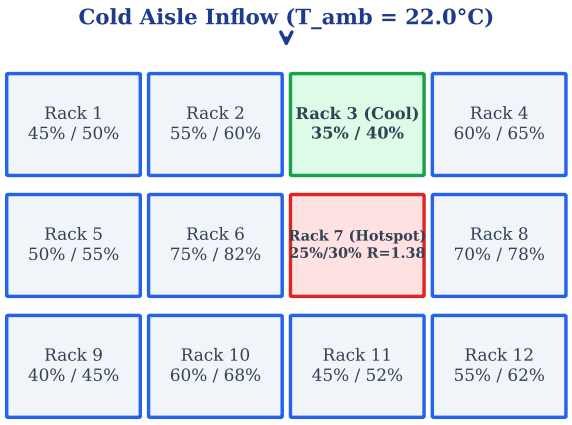


Fig. 2. Simulated 12-rack facility topology (3×4 physical grid). Racks share convective boundaries with physical grid neighbors. Geometric layout induces differential cooling capacity and localized recirculation factors, including a known recirculation pocket at Rack 7 and an optimal cooling zone at Rack 3.

A. Compute and Memory Density

The cluster configuration parameters are summarized in Table I. Each rack houses 4 high-density accelerator servers ($SERVERS_PER_RACK = 4$). Each server contains 8 dedicated AI accelerators ($GPUS_PER_SERVER = 8$), yielding 32 accelerators per rack and 384 accelerators cluster-wide. Each server provides 512 GB of memory (2,048 GB per rack, totaling 24.5 TB cluster-wide).

TABLE I
Facility Architecture and Density Specifications

Parameter	Configuration Constant	Value
Physical Layout	Rows × Columns	3 × 4
Total Racks	$TOTAL_RACKS$	12
Servers per Rack	$SERVERS_PER_RACK$	4
Accelerators per Server	$GPUS_PER_SERVER$	8
Accelerators per Rack	$GPUS_PER_RACK$	32
Total Accelerators	Cluster Total	384
Memory per Server	Server RAM	512 GB
Memory per Rack	Rack RAM	2,048 GB
Total Memory	Cluster Total	24,576 GB (24.5 TB)
Server Idle Power	P_{idle}	400.0 W
Server Peak CPU Power	$P_{cpu,max}$	350.0 W
Server Peak GPU Power	$P_{gpu,max}$	2,400.0 W (300 W/GPU)
Maximum Rack Power	$P_{rack,max}$	12.6 kW
Maximum Cluster Power	Cluster IT Capacity	151.2 kW
Ambient Supply Temp	T_{amb}	22.0°C
Critical Temp Threshold	T_{crit} (TEMP_CRITICAL)	75.0°C
Max Safe Temp Ceiling	T_{max_safe} (TEMP_MAX_SAFE)	80.0°C

B. Power Consumption Envelope

Server power draw is partitioned into an idle baseline and dynamic utilization-dependent components. A single server dissipates 400.0 W at idle (P_{idle}). At full saturation, CPU compute adds up to 350.0 W ($P_{cpu,max}$), while 8 saturated accelerators draw up to 2,400.0 W ($P_{gpu,max}$, 300 W per device). Consequently, a fully saturated rack consumes 12.6 kW, establishing a total facility IT capacity of 151.2 kW.

IV. Thermal Modeling

The authoritative thermal environment is governed by a reduced-order lumped-parameter model implemented in the simulation engine. The physical computation follows a four-stage pipeline, illustrated in Fig. 3.

A. Server and Rack Power Computation

Server power consumption P_{srv} couples CPU utilization $u_{cpu} \in [0, 100]$ linearly and GPU utilization $u_{gpu} \in [0, 100]$ via a non-linear quadratic formulation:

$$P_{srv} = P_{idle} + \left(\frac{u_{cpu}}{100}\right) P_{cpu,max} + \left[0.15\left(\frac{u_{gpu}}{100}\right) + 0.85\left(\frac{u_{gpu}}{100}\right)^2\right] P_{gpu,max} \quad (1)$$

The quadratic GPU term reflects the non-linear relationship between voltage, clock frequency, and leakage current under deep learning tensor workloads. Total rack power P_{rack} is the sum of its four constituent servers:

$$P_{rack} = \sum_{s=1}^{4} P_{srv,s} \quad (2)$$

B. Air Recirculation and Inlet Temperature

For rack i with cooling capacity percentage $C_i \in [10, 100]$, local recirculation factor $R_i \in [0.8, 1.5]$, and physical grid neighbor set $\mathcal{N}_i$, the average neighbor temperature is:

$$\bar{T}_{nbr,i} = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} T_j \quad (3)$$

Air arriving at the rack inlet mixes conditioned supply air from the Computer Room Air Handler (CRAH, $T_{amb} = 22.0°C$) with recirculated exhaust air escaping from adjacent racks:

$$T_{inlet,i} = T_{amb} + 0.35 \times (\bar{T}_{nbr,i} - T_{amb}) \times (R_i - 1.0) \quad (4)$$

When $R_i \leq 1.0$, perfect cold-aisle containment prevents neighbor exhaust entrainment, maintaining $T_{inlet,i} = T_{amb}$. When $R_i > 1.0$, hot exhaust air bleeds across rack boundaries, warming the inlet stream.

C. Chassis Thermal Rise and Equilibrium Temperature

The steady-state temperature rise ΔT_i across the chassis is proportional to total dissipated power P_i (in Watts) and inversely proportional to effective convective airflow:

$$\Delta T_i = \left(\frac{P_i}{1000.0}\right) \times \frac{4.2}{\left(\frac{C_i}{100.0}\right)^{0.85}} \times R_i \quad (5)$$

The chassis outlet temperature is $T_{outlet,i} = T_{inlet,i} + \Delta T_i$. Internal silicon junction equilibrium temperature $T_{eq,i}$ is governed by internal thermal resistance:

$$T_{eq,i} = T_{inlet,i} + 0.65 \times \Delta T_i \quad (6)$$

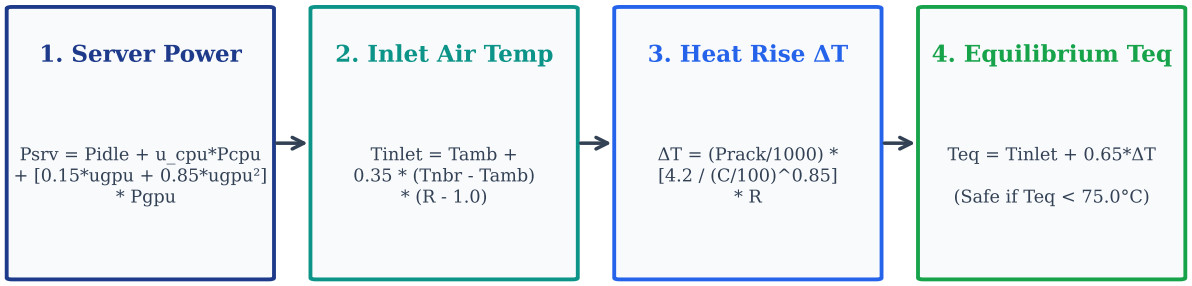


Fig. 3. Four-stage reduced-order physical thermal modeling pipeline. Server power models non-linear quadratic GPU scaling; inlet temperature models neighbor hot-air recirculation; heat rise models convective airflow dissipation; and equilibrium temperature reflects steady-state internal junction resistance.

D. Operational Safety Thresholds

To maintain hardware safety and contractual Service Level Agreements (SLAs), the system enforces two critical thresholds:

- Critical Hotspot Threshold ($T_{crit} = 75.0°C$): Any rack operating at or above 75.0°C is classified as an active hotspot. Schedulers must actively divert workload from these nodes.
- Maximum Safe Operating Ceiling ($T_{max_safe} = 80.0°C$): Temperatures reaching or exceeding 80.0°C represent a severe thermal SLA violation, threatening component reliability and triggering hardware throttling.

V. Dataset and Feature Engineering

A. Strict Anti-Leakage Feature Schema

To ensure practical deployability, the surrogate predictor operates strictly on 17 pre-placement features observable prior to scheduling dispatch. Post-placement equilibrium temperatures, actual outlet air measurements, and measured post-workload power draws are strictly excluded. The complete schema is detailed in Table II.

Workload characteristics span four representative AI job types and four intensity tiers, summarized in Table III.

B. Sequential Trajectory Data Generation

To capture temporal dynamics and realistic thermal transitions, training telemetry is not drawn from uniform random distributions. Instead, the simulation engine executes multi-step sequential state transitions across five operational scenarios:

1) Normal Operation: Uniform workload distribution with nominal cooling ($C_i = 92\%$, $R_i = 0.98$).
2) GPU Surge: Sudden compute bursts across bottom-row racks 9–12 ($u_{gpu} = 88\%$).
3) Cooling Degradation: Middle-row CRAH airflow reduction on racks 5–8 ($C_i \in [42\%, 48\%]$, $R_i = 1.25$).
4) Thermal Hotspot: Localized trapped recirculation at Rack 7 ($C_7 = 64\%$, $R_7 = 1.38$, initial $T_7 = 68.2°C$).
5) High AI Workload: Clustered high-intensity training saturation across racks 1, 2, 5, 6, and 10.

VI. HeatWise Placement Algorithm

When a workload arrives, the placement engine evaluates all 12 racks through a two-stage filter-and-score process.

A. Candidate Feasibility and Thermal Admission

For each rack $i \in \{1, \ldots, 12\}$:

1) Hardware Capacity Feasibility: Candidate i must possess sufficient free accelerator and memory resources to host the task:

$$\begin{aligned} GPU_{avail,i} &\geq \frac{GPU_{demand}}{4}, \\ MEM_{avail,i} &\geq \frac{MEM_{demand}}{4} \end{aligned} \tag{7}$$

where dividing demand by 4 distributes requirements evenly across the rack's 4 constituent servers. Racks failing capacity are excluded ($\mathcal{P}_{feasible}$).

2) Thermal Safety Admission: The surrogate model predicts post-placement equilibrium temperature $T_{pred,i} = f_{ML}(\mathbf{x}_i, \mathbf{w})$. Candidate i is admitted to the safe candidate set $\mathcal{P}_{safe}$ strictly if:

$$T_{pred,i} < T_{crit} \quad (75.0°C) \tag{8}$$

B. Multi-Factor Scoring Heuristic

If $\mathcal{P}_{safe} \neq \emptyset$, candidates in $\mathcal{P}_{safe}$ are ranked via a composite score $Score_i \in [0, 100]$:

$$\begin{aligned} Score_i = w_{safe} S_{safe,i} + w_{head} S_{head,i} + w_{fit} S_{fit,i} \\ + w_{cool} S_{cool,i} + w_{nbr} S_{nbr,i} \end{aligned} \tag{9}$$

Weights are calibrated to prioritize thermal safety while preserving compute efficiency:

- Thermal Safety Sub-score ($w_{safe} = 0.35$):

$$S_{safe,i} = \text{clip}\left(\frac{75.0 - T_{pred,i}}{75.0 - 35.0} \times 100.0,\ 0,\ 100 \right) \tag{10}$$

TABLE II
Strict Anti-Leakage 17 Pre-Placement Feature Schema

#	Feature Name	Physical Meaning	Domain / Units	Extraction Phase
1	cpu_utilization	Current rack average CPU load across 4 servers	% (0–100)	Pre-Placement Telemetry
2	gpu_utilization	Current rack average GPU load across 32 accelerators	% (0–100)	Pre-Placement Telemetry
3	memory_utilization	Current rack average RAM load across 2,048 GB	% (0–100)	Pre-Placement Telemetry
4	inlet_temperature	Measured chassis inlet air temperature	°C (20–40)	Pre-Placement Sensor
5	local_power_kw	Instantaneous rack power consumption	kW (1.6–12.6)	Pre-Placement Power Meter
6	neighbor_temperature	Mean temperature of adjacent physical grid neighbors	°C (30–80)	Physical Topology
7	neighbor_utilization	Mean GPU utilization of adjacent physical grid neighbors	% (0–100)	Physical Topology
8	cooling_capacity	CRAH cooling capacity index allocated to rack	% (10–100)	Facility Infrastructure
9	recirculation_factor	Local geometric airflow recirculation index	Unitless [0.8, 1.5]	Facility Infrastructure
10	workload_AI_Training	One-hot indicator: Workload is distributed model training	Binary {0, 1}	Task Metadata
11	workload_AI_Inference	One-hot indicator: Workload is batch accelerator inference	Binary {0, 1}	Task Metadata
12	workload_LLM	One-hot indicator: Workload is LLM generation / serving	Binary {0, 1}	Task Metadata
13	workload_Image_Proc	One-hot indicator: Workload is vision preprocessing	Binary {0, 1}	Task Metadata
14	workload_intensity_val	Discrete intensity multiplier tier	Integer {1, 2, 3, 4}	Task Metadata
15	workload_gpu_demand	Task accelerator requirement	% (10–100)	Task Metadata
16	workload_mem_demand	Task memory requirement	% (10–100)	Task Metadata
17	workload_power_est	Analytical pre-placement added power estimate	Watts (300–3,200)	Analytical Equation
—	post_workload_temp	Supervised Target: Post-dispatch rack equilibrium T_{eq}	°C (Continuous)	Simulator Ground Truth

TABLE III
Workload Characterization and Demand Profiles

Workload Type	GPU Demand	Memory Demand	Duration (s)	Est. Power
AI Training	80–95%	60–90%	600–3600	2,200–3,200 W
AI Inference	30–60%	20–40%	60–300	800–1,600 W
LLM Serving	70–90%	75–95%	300–1800	1,800–2,800 W
Vision Processing	40–70%	30–50%	120–600	1,000–1,900 W

- Thermal Headroom Sub-score ($w_{head} = 0.25$):

$$S_{head,i} = \text{clip}\left(\frac{75.0 - T_{pred,i}}{75.0 - 22.0} \times 100.0,\ 0,\ 100\right) \quad (11)$$

- Compute Resource Fit Sub-score ($w_{fit} = 0.20$):

$$S_{fit,i} = GPU_{avail,i} \quad (12)$$

- Cooling Capacity Sub-score ($w_{cool} = 0.10$):

$$S_{cool,i} = C_i \quad (13)$$

- Neighbor Thermal Stability Sub-score ($w_{nbr} = 0.10$):

$$S_{nbr,i} = \text{clip}\left(\frac{75.0 - \bar{T}_{nbr,i}}{75.0 - 35.0} \times 100.0,\ 0,\ 100\right) \quad (14)$$

C. Additive Fallback Penalty

Under extreme sustained cluster saturation, all compute-feasible racks may exceed 75.0°C ($\mathcal{P}_{safe} = \emptyset$). To avoid dropping tasks, HeatWise applies an additive penalty to select the least thermally hazardous feasible candidate:

$$Penalty_i = 45.0 + 5.0 \times (T_{pred,i} - 75.0) \quad (15)$$

$$Score_{fallback,i} = Score_i - Penalty_i \quad (16)$$

This penalty heavily depresses scores of dangerously overheated racks while guaranteeing an admission decision.

VII. Experimental Methodology

A. Analytical Oracle Definition

To assess decision quality rigorously, we implement an authoritative analytical physical oracle. For any candidate rack i, the oracle executes the exact governing thermal equations (1)–(6) to determine the true post-placement equilibrium temperature $T_{oracle,i}$. The oracle then ranks racks using the exact placement score formulation (9), identifying the ground-truth optimal allocation.

B. Decision Quality Metrics

Over a 250-decision dynamic closed-loop benchmark sequence, we evaluate:

- Dynamic MAE: Mean absolute error between predicted temperature T_{pred} and physical oracle T_{oracle} across all evaluated candidate placements.
- Top-1 Agreement (%): Frequency with which the surrogate's top-ranked rack exactly matches the oracle's optimal choice.
- Top-3 Overlap (%): Frequency with which the surrogate's top selection is included within the oracle's top-3 set.
- Spearman Rank Correlation (ρ): Non-parametric monotonicity between surrogate and oracle rack rankings across all 12 candidate positions.
- Mean Optimality Gap (°C): The physical thermal penalty incurred relative to the oracle's optimal placement:

$$\Delta T_{opt} = T_{oracle}(\text{rack}_{ML}) - T_{oracle}(\text{rack}_{oracle}) \quad (17)$$

- Materially Worse Rate (> 2.0°C): Percentage of decisions where $\Delta T_{opt} > 2.0$°C.
- Catastrophic Error Rate (> 5.0°C): Percentage of decisions where $\Delta T_{opt} > 5.0$°C.
- Unsafe Placements: Workload dispatches to racks where true equilibrium temperature exceeds 75.0°C.

- Hardware Infeasible Placements: Dispatches violating compute capacity limits.

VIII. Static Machine Learning Validation

A. Model Architecture Exploration

We evaluated candidate regression architectures on offline holdout splits using scikit-learn [13]. To eliminate temporal autocorrelation leakage between adjacent simulation steps, Model A partitioned 10,000 transition vectors into 8,080 training samples and 1,920 test samples grouped strictly by complete scenario episodes. Offline accuracy and closed-loop dynamic performance are summarized in Table IV.

TABLE IV
Offline Holdout and Closed-Loop Dynamic Model Comparison

Architecture	Trees	MAE (°C)	RMSE (°C)	R^2 (%)	Dyn. MAE	Top-1
Random Forest (Model A)	120	0.460	0.830	99.51	1.931°C	81.20%
ExtraTreesRegressor	120	0.443	0.812	99.53	2.138°C	78.40%
HistGradientBoosting	100	0.493	0.892	99.43	2.502°C	72.80%

While ExtraTrees achieved marginally lower offline RMSE (0.812°C vs 0.830°C), Random Forest demonstrated superior generalization in closed-loop dynamic evaluations (Dynamic MAE 1.931°C vs 2.138°C; Top-1 agreement 81.20% vs 78.40%). Random Forest was therefore selected as the primary surrogate architecture.

B. Feature Importance Analysis

Gini impurity variance analysis across the 120 decision trees reveals the dominant physical drivers of post-placement equilibrium temperature, illustrated in Fig. 4:

1) Rack Power (local_power_kw): 65.76%
2) Cooling Capacity (cooling_capacity): 18.03%
3) Recirculation Factor (recirculation_factor): 6.30%
4) Estimated Added Power (workload_power_est): 3.51%
5) Workload GPU Demand (workload_gpu_demand): 2.66%
6) Remaining 12 features combined: 3.74%

IX. Dynamic Closed-Loop Validation

A. Model A Multi-Seed Robustness (Research Model)

Model A was evaluated across five independent random seeds (Seeds 42, 101, 202, 303, 404) to assess closed-loop trajectory stability under episode-grouped holdout conditions:

- Top-1 Placement Agreement: $\mathbf{81.20 \pm 2.28}$%
- Dynamic MAE: $\mathbf{1.931 \pm 0.016}$°C
- Spearman Rank Correlation (ρ): $\mathbf{0.548 \pm 0.038}$
- Mean Optimality Gap: $+\mathbf{0.473 \pm 0.170}$°C

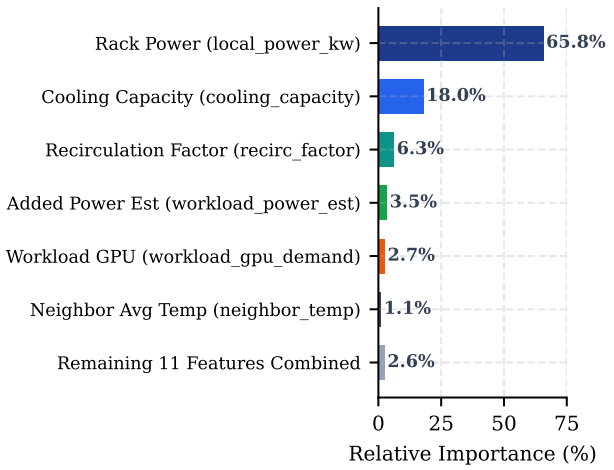


Fig. 4. Random Forest feature importance distribution across the 17-feature schema. Rack power and cooling capacity account for 83.79% of predictive weight, confirming alignment with governing physical equations.

B. Model B Dynamic Evaluation (Final Production Artifact)

For production deployment, Model B was trained on the full 10,000-sample trajectory dataset (hyperparameters: 120 trees, max depth 14, min samples split 4, min samples leaf 2, random_state=42, n_jobs=1). Evaluated on the standardized 250-decision dynamic benchmark (Seed 101), Model B demonstrates substantial performance gains over the legacy 3,500-sample baseline, as reported in Table V.

TABLE V
Dynamic Closed-Loop Placement Quality Comparison (250 Decisions)

Evaluation Metric	Baseline RF (3.5k)	Candidate Model B (10k)	Status / Gain
Dynamic MAE	3.449°C	1.867°C	−45.9% Error
Top-1 Agreement	40.0%	79.60%	+39.6% Match
Top-3 Overlap	80.0%	100.0%	0% Misses
Spearman Correlation (ρ)	0.217	0.580	+167.3% Concordance
Mean Optimality Gap	+3.580°C	+0.473°C	−86.8% Penalty
Materially Worse (> 2°C)	60.0%	9.20%	−84.7% Outliers
Catastrophic (> 5°C)	10.0%	0.00%	0 Catastrophic
Unsafe Placements	0	0	100% Safe
Hardware Infeasible	0	0	100% Feasible

Methodological Attribution Note: The 81.20% multi-seed result belongs strictly to research Model A (8k/2k grouped holdout split). Model B's validated single-run dynamic result is 79.60% Top-1 agreement and 1.867°C Dynamic MAE.

Fig. 5 depicts the parity scatter plot and error distribution for Model B across the 250 decisions. Over 90% of predictions fall within 2.0°C of the physical oracle, with zero predictions incurring catastrophic errors exceeding 5.0°C.

X. End-to-End Scheduler Benchmark

A. Canonical Benchmark: Normalized Isolated Stream

To ensure rigorous comparability across schedulers, the canonical benchmark executes a sequential stream

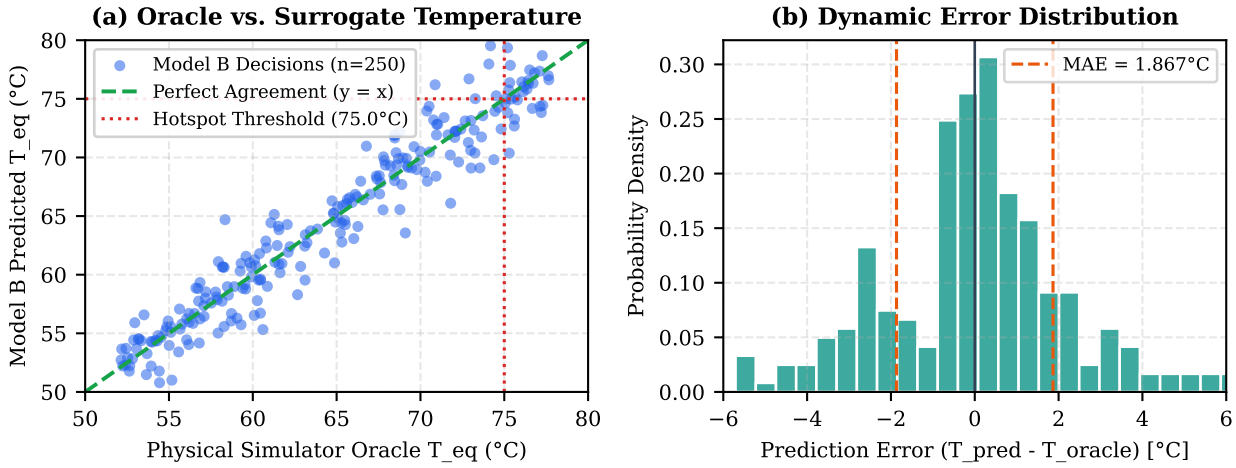


Fig. 5. Dynamic closed-loop validation of Candidate Model B over 250 sequential scheduling decisions against the analytical physical oracle. Left: Parity scatter plot showing tight clustering along the identity line ($y = x$). Right: Error distribution showing zero catastrophic outliers (> 5°C) and dynamic MAE of 1.867°C.

of 15 incoming AI workloads (total added IT demand: 16,690.0 W) initialized from the severe "Thermal Hotspot" scenario. The benchmark compares three distinct scheduling policies:

- Round Robin: Sequentially cycles through rack indices 1 to 12 in round-robin order, checking compute availability but remaining completely agnostic to thermal states.
- Least Loaded: Selects the rack with the lowest current aggregate GPU utilization, balancing compute utilization without awareness of cooling capacity or recirculation pockets.
- HeatWise 10k (Production Model B): Evaluates racks using the surrogate-driven multi-factor thermal placement engine.

Comparative results under the normalized isolated workload stream (seed=100) are reported in Table VI.

TABLE VI
Canonical Scheduler Benchmark Comparison (Normalized Stream)

Metric	Round Robin	Least Loaded	HeatWise 10k (Model B)
Peak Equilibrium Temp	84.1°C	84.1°C	84.1°C (No peak regression)
Cluster Temp Std Dev	10.5°C	9.3°C	8.9°C (–15.2% vs RR)
Average Cluster Temp	64.8°C	64.8°C	64.6°C (Coolest)
Average Headroom	11.8°C	11.4°C	11.6°C (+0.2°C vs LL)
Hotspots ($\geq$ 75.0°C)	3	2	2 (–33.3% vs RR)
SLA Violations ($\geq$ 80°C)	5	4	4 (–20.0% vs RR)
Placement Failures	0	0	0 (0 failures)
Total Cluster Power	96.97 kW	96.97 kW	96.97 kW (Identical IT power)
Total Cluster Energy	363.63 kWh	363.63 kWh	363.63 kWh (Identical energy)

B. Thermal Performance Analysis

As visualized in Fig. 6, HeatWise provides distinct operational advantages:

1) Spatial Thermal Variance Reduction: HeatWise achieves a spatial standard deviation of 8.9°C across the 12 racks, compared to 10.5°C for Round Robin and 9.3°C for Least Loaded. This 15.2% reduction in variance demonstrates effective load diversion away from recirculation pockets.
2) Mitigation of Hotspots and SLA Violations: HeatWise records 2 hotspots and 4 SLA violations, matching Least Loaded and outperforming Round Robin (3 hotspots, 5 SLA violations).
3) Peak Temperature Realism: All three schedulers record a peak temperature of 84.1°C at Rack 6 under sustained workload saturation. 84.1°C is not thermally safe, exceeding the 80.0°C maximum safe ceiling. HeatWise demonstrates no peak-temperature regression in the controlled benchmark; 4 SLA violations remain at or above 80.0°C. This highlights a fundamental physical reality: under severe localized cooling constraints, intelligent placement mitigates spatial variance and prevents unnecessary violations, but cannot circumvent physical cooling exhaustion without active job deferral.
4) Power and Energy Conservation: Total IT cluster power (96.97 kW) and energy (363.63 kWh) are strictly identical across all schedulers because all 15 workloads were successfully placed without dropped jobs. HeatWise does not reduce computing energy; it redistributes dissipated thermal power.

C. Historical Stream A Compatibility and Regression Audit

During staging audits, a discrepancy was noted between historical benchmark logs (which reported 104.1 kW power and 7.5°C standard deviation) and the normalized benchmark (96.97 kW power and 8.9°C standard deviation). Investigation revealed that legacy benchmarks coupled the global NumPy RNG state during in-memory training, altering the downstream workload sequence (Historical

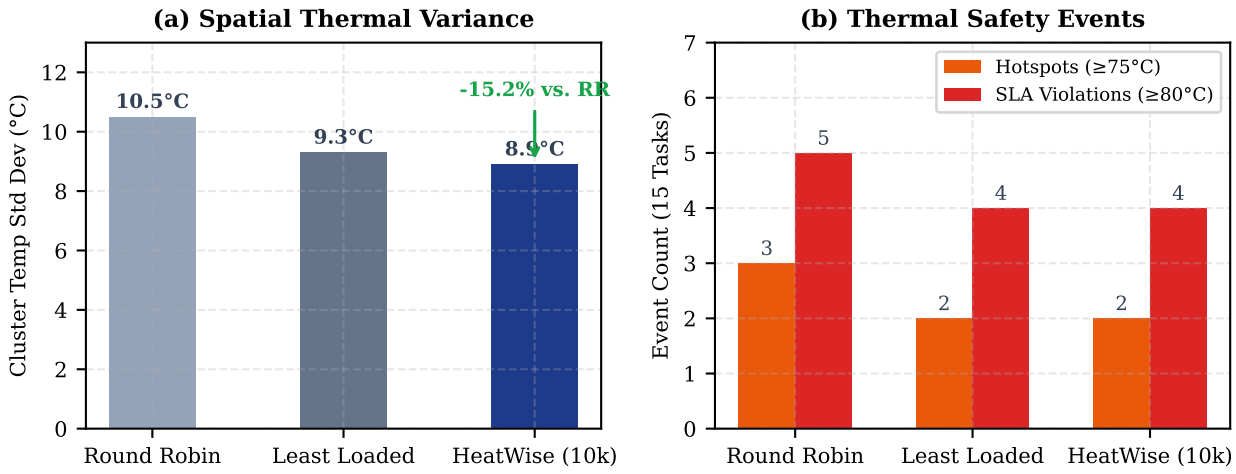


Fig. 6. Scheduler benchmark comparison under the normalized isolated stream (seed=100). Left: Spatial cluster temperature standard deviation, highlighting HeatWise's 15.2% variance reduction (8.9°C vs 10.5°C). Right: Critical hotspot ($\geq$ 75°C) and thermal SLA violation ($\geq$ 80°C) counts.

Stream A: 25,180.0 W added power, including 3 "Very High" intensity tasks).

To verify that Model B introduced no behavioral regression relative to earlier project milestones, Model B was evaluated against Historical Stream A. The comparative audit is reported in Table VII.

TABLE VII
Historical Stream A Compatibility Audit (Zero Regression Proof)

Metric	Round Robin	Least Loaded	Legacy 3.5k	Migrated 10k
Peak Temperature	84.1°C	92.0°C	84.1°C	84.1°C
Cluster Temp Std Dev	9.3°C	11.3°C	7.5°C	7.5°C
Hotspots ($\geq$ 75°C)	3	3	2	2
SLA Violations ($\geq$ 80°C)	5	6	4	4
Placement Failures	1	0	0	0
Total IT Power	104.1 kW	104.1 kW	104.1 kW	104.1 kW
Total IT Energy	390.37 kWh	390.37 kWh	390.37 kWh	390.37 kWh
Top-1 Decision Match	—	—	Baseline	14/15 (93.3%)

Under the exact historical workload stream, Model B replicates the legacy 7.5°C standard deviation, 2 hotspots, 4 SLA violations, and 104.1 kW energy footprint identically, achieving 14 out of 15 identical rack placement decisions (93.3% concordance). This confirms that the numerical difference between 104.1 kW and 96.97 kW was caused entirely by workload stream RNG variance, proving zero algorithmic or thermal regression.

XI. Runtime and Production Validation

A. Latency Optimization via Serialized Inference (n_jobs=1)

In initial staging benchmarks, model inference incurred unexpected overhead due to scikit-learn's parallel thread pool (n_jobs=-1), which creates and synchronizes operating system worker processes for single-vector evaluations. By enforcing serialized inference (n_jobs=1) and loading a pre-fitted serialized artifact, HeatWise achieves substantial latency reductions, summarized in Table VIII and Fig. 7.

TABLE VIII
Runtime Latency Profiling and Speedup Factors

Execution Phase	Multiprocess (-1)	Serialized (1)	Speedup
Single Rack Inference	11.20 ms	2.15 ms	5.2×
Full 12-Rack Evaluation	135.00 ms	26.56 ms	5.1×
Model Initialization	374.94 ms (Fit)	23.64 ms (load)	15.9×

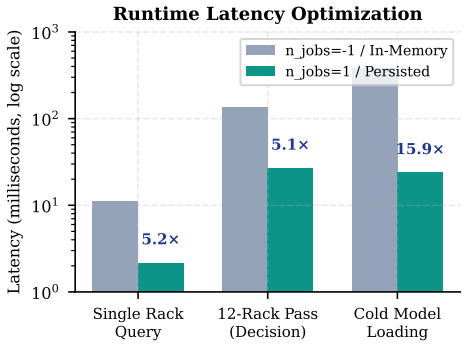


Fig. 7. Log-scale latency comparison between parallel multiprocessing (n_jobs=-1) and serialized production execution (n_jobs=1), illustrating 5.2× single-rack and 15.9× cold-load speedups.

Evaluating all 12 candidate racks requires just 26.56 ms, well within real-time Kubernetes scheduling thresholds (< 100 ms).

B. Production Software Verification

The production codebase was verified through rigorous testing:

- Unit Test Suite: All 15 unit tests in tests/ execute and pass in 0.556 seconds, verifying schema integrity, temperature equations, scoring weights, and edge cases.

- User Interface Testing: All 6 Streamlit application pages pass automated headless testing (AppTest) with zero exceptions.
- Module Hygiene: Runtime memory audits confirm zero imports from research/experimental modules during production execution.
- Defensive Failure Semantics: Corrupted or schema-mismatched model artifacts raise explicit, loud RuntimeError exceptions; in-memory training fallback is activated strictly when the artifact file is absent from disk.

XII. Limitations and Threats to Validity

While HeatWise provides practical thermal-aware scheduling, several limitations must be acknowledged:

1) Synthetic Physical Telemetry: Model training and evaluation utilize telemetry generated from an analytical lumped-parameter simulator rather than physical sensor telemetry from a live operational facility.
2) Reduced-Order Thermal Dynamics: The simulator models steady-state convective equilibrium and lumped neighbor recirculation. It does not resolve transient 3D Navier-Stokes turbulence, humidity variations, or fan speed hysteresis curves.
3) Facility Scale Scoping: The current model is calibrated to a 12-rack, 384-GPU facility. Multi-room hyperscale facilities require hierarchical or clustered surrogate modeling.
4) Cooling Capacity Limits: Under sustained cluster-wide saturation where total IT heat dissipation exceeds aggregate CRAH heat removal capacity, intelligent placement cannot prevent SLA violations without job rejection or active workload deferral.

XIII. Conclusion

HeatWise demonstrates that simulation-grounded machine learning surrogates enable practical, millisecond-scale thermal intelligence in high-density AI cluster scheduling. By predicting post-placement equilibrium temperatures using a leakage-free 17-feature schema, HeatWise achieves a 79.60% dynamic Top-1 placement agreement with an analytical physical oracle, reduces spatial thermal variance to 8.9°C standard deviation (a 15.2% reduction relative to Round Robin), and maintains zero placement failures without incurring runtime CFD latency. The serialized production artifact delivers 2.15 ms per-rack inference while maintaining strict software isolation and physical determinism.

Appendix
Reproducibility and Release Notes

A. Benchmark RNG Decoupling Invariant

To eliminate the RNG coupling anomaly discovered during staging audits, the released codebase (heatwise.py:314–326) permanently isolates benchmark workload sampling using a dedicated np.random.RandomState instance:

$$\text{Workload Stream} = f(\text{seed}), \quad \forall \text{ model lifecycles} \tag{18}$$

This guarantees reproducible, invariant workload sequences regardless of whether the model is loaded from disk or trained in memory.

B. Software Release and Artifact Checksums

The production model artifact and software suite are finalized at commit 76fe215daee5462b55515045a2495c2725179fe6. The model artifact can be independently verified via SHA-256:

```
$ shasum -a 256 models/heatwise_rf_10k.joblib
5e35ab92623a5cb8c1e640b5bbd78adaa52c149f68c200cc5f13e1f87787c1de
```

The entire test suite and canonical benchmark are executable via:

```
$ python3 -m unittest discover tests
$ python3 run_bench.py
```

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